

# Dylan Dsouza

San Diego, California

✉ [dydsouza@ucsd.edu](mailto:dydsouza@ucsd.edu)

🌐 [dylansouza.com](https://dylansouza.com)

🌐 [linkedin.com/in/dsouza-dylan](https://www.linkedin.com/in/dsouza-dylan)

🐙 [github.com/dsouza-dylan](https://github.com/dsouza-dylan)

## EDUCATION

**Bachelor of Science, Data Science** | University of California San Diego | GPA: 3.95 – Provost Honors Expected Mar 2027  
**Relevant Coursework:** Applied Data Science, Machine Learning Theory, Recommender Systems, Statistical Inference  
**Activities and Societies:** Hedge Fund Club, Data Science Student Society, Peaks and Professors

## TECHNICAL SKILLS

**Languages:** Python, SQL, R, Java, C, C++, HTML, CSS, JavaScript

**Libraries/Frameworks:** NumPy, pandas, scikit-learn, PyTorch, TensorFlow, OpenCV, Plotly, LangChain, Jupyter, Git/GitHub

**Core Competencies:** Data Cleaning/Visualization, EDA, Web Scraping, A/B Testing, Financial Modeling, Time Series Analysis

## EXPERIENCE

### Machine Learning Researcher

Oct 2025 – Present

*Engineers for Exploration*

*San Diego, CA*

- Engineered ensemble machine learning models (70% mIoU baseline) for automated mangrove mapping, developing end-to-end image segmentation pipelines to process satellite and drone imagery for pixel-level land-cover classification.
- Integrated 7 geospatial datasets (SpaceNet, xView, LandCover.ai) to optimize model architecture and feature engineering, accelerating ML-assisted labeling workflows across 4 environmental classes (road, building, water, mangrove).

### AI Research & Development Intern

Oct 2025 – Present

*Computer Science & Engineering Society*

*San Diego, CA*

- Building AgentSpace, an AI platform leveraging GraphRAG architecture and game-theoretic personality modeling to model professional decision-making patterns, with initial focus on prototyping intelligent agentic tutoring systems.
- Developed retrieval agent prototype using LangChain and RAG pipelines within cross-functional team of 9, experimenting across 6 Gemini model configurations to benchmark performance and optimize response generation workflows.

### Research Data Analyst

May 2025 – Present

*Scripps Institution of Oceanography*

*San Diego, CA*

- Processed and annotated 9,000+ georeferenced coral imagery points using in-house photogrammetry software (Viscore), maintaining structured datasets for downstream temporal analysis across 15 reef sites on Palmyra Atoll.
- Executed time series analysis across 16 time points (2014-2025) quantifying impact of depth and 5 treatment variables on reef health metrics, delivering insights on coral bleaching trends to inform data-driven restoration strategies.

### Quantitative Analyst

Oct 2025 – Dec 2025

*Sustainable Investment Group*

*San Diego, CA*

- Implemented automated Python data pipeline using FinBERT NLP, Yahoo Finance API, and Tavily to generate trading signals by quantifying lagged correlations between 25 weeks of news sentiment and 5+ market KPIs (returns, volatility, volume).
- Co-led quantitative investment thesis for Under Armour, integrating ARIMA time series forecasting, sentiment analysis, and DCF financial modeling to deliver data-driven sell recommendation in competitive 5-team stock pitch.

### Market Research Intern

Jul 2023 – Aug 2023

*Bennett Coleman & Co. Ltd. (The Times of India)*

*Mumbai, India*

- Conducted quantitative research on 56,000+ newspaper headlines using TF-IDF vectorization and multi-output linear regression, training a model to predict relevance scores across 5 key market segments.
- Delivered 5-category classification prototype achieving 85% accuracy, enabling automated headline categorization across Business, Politics, Technology, Social Issues, and Lifestyle segments for editorial workflows.

## PROJECTS

### NextMove: Predicting Next Workout Activity

Nov 2025 – Dec 2025

*Course Project – Recommender Systems*

[View Project](#)

- Engineered temporal, behavioral, and physiological features from 20,000+ sequential workout records in the Endomondo dataset, designing transition-based training instances to predict next activity type without look-ahead bias.
- Compared heuristic baselines (frequency, recency, weighted) against supervised ML models (Logistic Regression, Naïve Bayes, SVM, Random Forest), achieving 87% accuracy with Random Forest using chronological splits and balanced class weighting.

### ENSOCast: Interactive Climate Analytics Dashboard

May 2025 – Oct 2025

*Personal Project (Presented at San Diego Undergraduate Tech Conference 2025)*

[View Project](#)

- Built a Streamlit dashboard with interactive Plotly visualizations for exploring 40+ years of ocean temperature data and ENSO phases, enabling users to train and compare custom ML models.
- Developed and evaluated advanced ML models, including Random Forest, XGBoost, 1D CNN, LSTM, and Ensemble Learning, using temporal, seasonal, and interaction-based features, achieving up to 82% accuracy in ENSO phase prediction.